

Aditya Hansoty

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EDUCATION

University of Michigan

Bachelor of Science, Biomedical Engineering, Minor in Computer Science

Ann Arbor, MI

Aug. 2025 – May 2029

- GPA: 3.73/4.0
- Activities: Biomedical Engineering Society (Professional Development Chair), M-HEAL Team PACT (Design Subteam)

WORK EXPERIENCE

Systems Engineering Intern

Naval Medical Research Unit, Department of the Navy

Oct. 2026 – Dec. 2026

San Antonio, TX

- Processing physiological signals and applying deep learning to support medical device development for military medicine

Prosthetics Research Intern

Rehabilitation Robotics Lab, University of Illinois Chicago

May 2026 – Sept. 2026

Chicago, IL

- Designed and iterated threaded encoder caps for a powered ankle-foot prosthesis in SolidWorks, progressing through 3D-printed and CNC-machined revisions to improve mechanical fit
- Controlled a powered ankle-foot prosthesis via TwinCAT and led Vicon motion capture across 10 human-subject trials; built Python pipelines synchronizing EMG, ECG, and metabolic cost data for a manuscript presented at IROS 2026
- Engineered a signal-agnostic AR biofeedback platform (Python, Unity, HoloLens 2) pixel-streaming live via WebSockets for real-time gait feedback; sole developer, open-sourced
- First-author manuscript submitted to ICRA 2027 demonstrating HoloLens 2 latency (120.2 ms) matched a wired monitor (119.2 ms) via OpenCV tracking; 2-subject pilot showed AR guidance reduced step-length asymmetry by 41% and 53%

Exoskeleton Research Intern

Locomotor Control Systems Laboratory, University of Michigan

Sept. 2026 – Present

Ann Arbor, MI

- Collecting and processing motion-capture and EMG data from clinical trials evaluating a wearable knee exoskeleton in patients with osteoarthritis and stroke
- Redesigning hip-belt geometry and dorsal motor-housing mounts in SolidWorks to improve wearability and reduce sag

Cancer Biology Research Intern

University of Chicago Medicine

June 2025 – Aug. 2025

Chicago, IL

- Investigated β -tubulin isotype knockouts in bone-metastatic breast cancer cells using cell culture, adhesion assays, Western blotting, and confocal microscopy; presented findings to 100+ researchers at the UChicago Cancer Center Symposium

PROJECT EXPERIENCE

TapTrack PD | MHacks 2026, 1st Place Tiger Data Track

Oct. 2026

- Built a wrist wearable on a FREE-WiLi for Parkinson's, using its accelerometer, mic, and buttons for an MDS-UPDRS-based motor check with FFT tremor detection and ElevenLabs voice guidance
- Stored checks and doses as TimescaleDB time-series data via FastAPI, streaming results to a clinician dashboard in ~1 s
- Integrated Fetch.ai, Gemini, Photon, and FinchNode to generate clinician reports, text caregivers, and format FHIR records

HPV Screening Device (M-HEAL) | SolidWorks

Sept. 2026 – Present

- Redesigning a urine-collection device in SolidWorks for point-of-care cervical cancer screening in Ghana, adapting geometry for pit and pan latrines
- Translating clinician and patient feedback from field interviews into design requirements for a low-cost device

Tibial Locking Compression Plate | SolidWorks, FEA, Simulink

Aug. 2026 – Sept. 2026

- Designed a parametric 10-hole Ti-6Al-4V locking compression plate in SolidWorks for tibial fracture fixation
- Ran static linear FEA in SolidWorks Simulation with ASTM F382 knife-edge pin/roller supports and inner-span loading, achieving factors of safety of 1.73 (500 N bending) and 2.36 (10 N·m torsion); mesh converged within 3%
- Developed a Simulink bone-healing model predicting a 46% reduction in plate stress over 12 weeks

EMG-Controlled Prosthetic Gripper | SolidWorks, Arduino, Python

May 2025 – July 2026

- Designed and 3D-printed a 2-finger prosthetic gripper integrating an MG996R servo, MyoWare 2.0 EMG sensor, and Arduino Uno for real-time gesture-controlled actuation
- Benchmarked two EMG classification pipelines, achieving 99.4% three-gesture classification accuracy using LDA

TECHNICAL SKILLS

Engineering: SolidWorks, SolidWorks Simulation (FEA), 3D Printing, CNC Machining, Arduino, Simulink, TwinCAT, Vicon Motion Capture, Signal Processing, Lab Streaming Layer

Programming: Python, C++, MATLAB, Unity (C#)

Data & ML: OpenCV, WebSockets, FastAPI, PostgreSQL/TimescaleDB, NumPy, Pandas, SciPy, scikit-learn